

Categorization of the hemodynamic response to hemodialysis: The importance of baroreflex sensitivity

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Abstract

Intradialytic hypotension (IDH) remains an important cause of morbidity and mortality in hemodialysis (HD) patients. The baroreflex arc is under autonomic control and regulates blood pressure. This study aimed to investigate the contribution of impaired baroreflex sensitivity (BRS) to the pathophysiology of IDH. Thirty-four chronic HD (12 IDH-prone, 22 IDH-resistant) patients underwent BRS measurement during HD with relative blood volume monitoring. During analysis, patients were separated into four age-matched groups according to resting BRS ≥ 4.5 ms/mmHg and hemodynamic stability. Resting BRS was extremely heterogeneous (geometric mean BRS 5.78 ± 1.41 [range 1.76–41.41] ms/mmHg). Relative blood volume reduction was well matched in all groups (mean reduction in relative blood volume for all patients $-6.74\% \pm 0.86\%$, $P > 0.05$). Thirty-seven episodes of IDH occurred in the IDH prone, reduced BRS group. Patients with impaired resting BRS and prone to IDH had markedly different responses to HD as compared to the preserved BRS group, but the total peripheral resistance response was significantly lower than in the IDH-resistant patients ($15.9\% \pm 2.1\%$ vs. $42.4\% \pm 3.0\%$, respectively, $P < 0.001$). In those patients prone to IDH and with impaired resting BRS, percentage reduction in cardiac output at the end of HD highly correlated with reduction in relative blood volume ($r=0.94$, $P=0.006$). Hypotension during dialysis may be an important source of recurrent cardiac injury and early recognition of those patients prone to relative symptomatic and asymptomatic hypotension remains important. Impaired resting BRS and recognition of a suboptimal peripheral pressor response, appear to predict those patients most likely to undergo hemodynamic instability and may assist in the pursuit of this elusive goal.

Key words: Blood pressure, autonomic dysfunction, heart rate variability, intradialytic hypotension, relative blood volume, baroreceptors

INTRODUCTION

Despite major technical advances in hemodialysis (HD), the incidence of intradialytic events has not decreased in recent years.¹ HD patients experience an enormous excess of cardiovascular (CV) morbidity and mortality.

This appears to be largely driven by factors that are different from those traditional characteristics observed in the nonuraemic general population.² Recent studies have identified that intradialytic hypotension (IDH) is a predictor of increased mortality.^{3,4} Chronic HD patients are already primed by a large number of structural and functional peripheral vascular and cardiac abnormalities to experience demand myocardial ischemia.⁵ HD is capable of inducing myocardial ischemia⁶ and recurrent ischemic insults lead to functional and structural myocardial change, eventually resulting in heart failure.^{7–9}

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Heart failure is associated with sympathetic overactivity and impaired baroreflex sensitivity (BRS)^{10,11} (an accepted marker of autonomic dysfunction). Modifications of the HD process to improve hemodynamic tolerability, have been shown to reduce the perturbation of myocardial blood flow and functional evidence of dialysis-induced ischemia.^{12,13} Identifying predictors of IDH and aberrant hemodynamic response during HD is therefore potentially important.

A variety of factors associated with the development of IDH have previously been reported. These include insufficient refill, left ventricular dysfunction, imbalance of vasoactive agents,^{14,15} excessive reduction in relative blood volume (RBV),¹⁶ and autonomic dysfunction.^{17–23} However, neither the relative contributions of these various putative mechanisms or the role of autonomic dysfunction have been fully elucidated.

Autonomic dysfunction and in particular, abnormal regulation of blood pressure (BP) through the baroreflex arc is well-recognized among HD patients.²⁴ Preserved sensitivity of this system is essential in the maintenance of normal CV function and treatment tolerability during dialysis. Reduced resting BRS is known to be an independent predictor of death in patients with and without chronic kidney disease.^{25,26}

Low-pressure stretch receptors located in the atria, ventricles, and pulmonary vessels are components of another important autonomic regulatory system known as the cardiopulmonary reflex. These receptors are activated by cardiac distension due to an increase in venous return, and elicit a sympathetic response. Mechanoreceptors within the left ventricular wall respond to an increase in end-diastolic pressure and result in reflex bradycardia and vasodilatation (Bezold–Jarisch reflex). All these reflexes are important in the maintenance of BP at a set-point. Scant knowledge exists regarding the heterogeneity of autonomic dysfunction during HD. Attempts to classify resting BRS and whether resting BRS can predict outcome has not been investigated previously.

The objective of this study was to investigate the relative impact of BRS values in response to RBV reduction during stable and unstable HD, and aimed to identify individual patterns of hemodynamic response on the basis of resting BRS.

METHODS

Patients and experimental schedule

Thirty-four chronic HD patients, who dialyze for 4 hours thrice weekly, were recruited randomly from a single

main hospital dialysis unit. Patient characteristics are presented in Table 1. IDH was classified as systolic blood pressure (SBP) ≤ 100 mmHg, even in the absence of symptoms, or a fall in SBP $> 10\%$ of the predialysis reading in association with any symptom classically associated with hypotension. Patients were classified as IDH-prone patients if they had experienced episodes of IDH in $> 30\%$ of HD sessions in the month before recruitment to the study.

Dry weight was established through clinical assessment and RBV monitoring before entry to the study. Each patient underwent 30 minutes of BRS monitoring whilst seated quietly in the dialysis chair immediately before receiving HD. Patients were divided into 4 groups for subsequent analysis according to a resting BRS of ≥ 4.5 ms/mmHg (the median value and indicative of preserved and impaired BRS, respectively), and hemodynamic instability. Group 1 patients were IDH resistant with reduced BRS; group 2 patients were IDH resistant with preserved BRS; group 3 were IDH prone with reduced BRS and group 4 patients were IDH prone with preserved BRS. No changes were made to individual HD prescriptions. Any vasoactive medications were left unchanged. Patients were only excluded if they had arrhythmias (incompatible with Finometer[®] analysis, Finapres Medical Systems, Arnhem, the Netherlands) or if they had a history of dialysis-induced angina. Appropriate written consent was obtained and the study was approved by the Derbyshire Local Research Ethics Committee.

HD schedule

Hemodialysis was performed using Hospal Integra[®] (Mirandola, Italy) dialysis monitors. Ultrapure dialysate was generated using water having undergone dual pass treatment. Patients were dialyzed using low-flux polysulphone dialyzers from 1.8 to 2.0 m² as per their usual prescription (LOPS[®] 18–20, Braun Medical Ltd, Sheffield, UK), bicarbonate buffering, a dialysate sodium concentration of 140 mmol/L, calcium concentration 1.25 mmol/L, and dialysate flow was 500 mL/min. No variations to individual blood pump speeds occurred throughout the study.

Dialysis prescriptions were held on a separate server and downloaded to the patient for each treatment. All treatments were monitored for changes in RBV by continually measuring hemoglobin concentration (HemoScan[®]), an integral feature in each Hospal Integra[®] monitor. All data pertaining to the dialysis session were uploaded to patient-specific files at the end of each treatment for subsequent analysis.

Table 1 Demographic data for all patients separated according to IDH resistant with impaired resting BRS (group 1); IDH resistant and preserved resting BRS (group 2); IDH prone and impaired resting BRS (group 3) and IDH prone and preserved resting BRS (group 4)

	Overall (n=34)	Stable (n=22)		Unstable (n=12)		P value
		Group 1 (n=10)	Group 2 (n=12)	Group 3 (n=6)	Group 4 (n=6)	
Age (years)	64.2 ± 2.0	61.8 ± 3.1	64.0 ± 3.8	59.2 ± 4.6	73.5 ± 4.3	0.159
Male (%)	25 (74)	7 (70)	10 (83)	3 (50)	5 (83)	0.447
HD vintage (months)	39.7 ± 7.2	25.6 ± 6.2	36.7 ± 9.2	21.8 ± 10.6	25.5 ± 7.25	0.611
Smoking history (%)	12 (30)	3 (30)	6 (50)	1 (17)	2 (30)	0.536
Fistulae (%)	28 (82)	7 (70)	11 (92)	4 (67)	6 (100)	0.254
Number of patients with Diabetes (%)	10 (29)	3 (30)	2 (17)	4 (67)	1 (17)	0.143
Number of patients with IHD (%)	9 (26)	1 (10)	6 (50)	0	2 (33)	0.068
Number of patients with PVD (%)	10 (29)	4 (40)	2 (17)	2 (33)	2 (33)	0.667
Number of patients with HT (%)	21 (62)	7 (70)	8 (67)	3 (50)	3 (50)	0.774
Angiotensin-converting enzyme inhibitors (%)	6 (18)	1 (10)	3 (25)	2 (33)	1 (17)	0.686
Calcium channel blockers (%)	8 (24)	2 (20)	4 (33)	2 (33)	2 (33)	0.604
β-blockers (%)	6 (18)	2 (20)	3 (25)	0	1 (17)	0.621
α-blockers (%)	2 (6)	0	0	2 (33)	0	0.081
Angiotensin receptor blockers (%)	2 (6)	0	0	2 (33)	0	0.019
Centrally acting antihypertensives (%)	1 (3)	0	1 (8)	0	0	0.596
Corrected calcium (mmol/L)	2.47 ± 0.03	2.51 ± 0.1	2.50 ± 0.04	2.35 ± 0.08	2.44 ± 0.02	0.300
Phosphate (mmol/L)	1.74 ± 0.09	1.61 ± 0.04	1.83 ± 0.19	2.07 ± 0.24	1.41 ± 0.19	0.148
Parathyroid hormone (nmol/L)	334.7 ± 54.8	271.8 ± 46.7	476.8 ± 135.2	299.2 ± 94.5	190.7 ± 52	0.263
Albumin (mmol/L)	33.0 ± 0.6	30.7 ± 1.13	33.17 ± 1.21	33.7 ± 1.02	35.7 ± 0.56	0.055
Kt/V	1.26 ± 0.04	1.25 ± 0.07	1.21 ± 0.07	1.36 ± 0.1	1.30 ± 0.12	0.641

Data presented as mean ± SEM.

HD=hemodialysis; HT=hypertension; IHD=ischemic heart disease; PVD=peripheral vascular disease.

Measurement of hemodynamic variables and BRS

Noninvasive continuous pulse wave analysis was used to determine hemodynamic variables and to derive measurements of BRS. The Finometer[®] utilizes photoplethysmography²⁷ to derive a pulse wave from a finger pressure cuff containing an infrared optical source and detection unit. The pulse wave is then subject to mathematical analysis based on the transport characteristics of peripheral blood vessels to reconstruct a central pressure waveform. This in turn is analyzed to calculate a wide variety of hemodynamic variables including SBP, diastolic BP (DBP), mean arterial pressure (MAP), heart rate (HR), cardiac output (CO), stroke volume (SV), and total peripheral resistance (TPR) on a beat-to-beat basis.^{28,29} All data were subsequently downloaded to a personal computer-based analysis program (Beatscope[™], Finapres Medical Systems), allowing averaging of results over defined time periods

and display of variables as percentage change from baseline. This technology provides unprecedented resolution of changes in the critical CV variables.²⁸ Continuous measurement of the interbeat interval (IBI) and BP by the Finometer[®] enables BRS to be calculated. Three consecutive changes in the R-R interval in the same direction are required before a phase shift calculation (incorporated into the software) is performed and therefore BRS can be measured noninvasively with accuracy and precision.³⁰ Baroreflex sensitivity software was available to us as a stand-alone DOS-based offline analysis tool by Karl Wesseling (TNO instruments, Amsterdam, the Netherlands).

STATISTICAL ANALYSIS

Data were analyzed with Prism v3 software (GraphPad Software, Inc., La Jolla, CA, USA). Hemodynamic and BRS data were analyzed in 5-minute averaged sections

throughout HD. Results are presented as mean $\pm$ SEM unless otherwise stated. Differences in BRS were analyzed with the Mann-Whitney test. Patient characteristics were analyzed with Fisher's exact test where appropriate. One-way ANOVA with Bonferroni's correction was used to analyze the demographic and hemodynamic variables between the groups. Pearson's correlation was used to assess degree of correlation between %CO change and RBV reduction after 240 minutes of dialysis. IDH episodes were analyzed with the Mann-Whitney test. P values <0.05 were considered significant.

RESULTS

Patient characteristics

The groups were well matched for age, gender, duration of HD, previous medical history including diabetes, routine hematology/biochemistry and vasoactive medication (apart from angiotensin II receptor blockers (Table 1). Overall, patients had a considerable range in resting BRS (1.76–41.1 ms/mmHg, Table 2). Mean RBV reduction was well matched in all groups ($-6.74\% \pm 0.86\%$, $P>0.05$). Appropriately, there were no episodes of IDH in the IDH-resistant group.

Hemodynamic changes during HD in IDH-resistant patients

There were marked differences in hemodynamic response to HD between those with impaired and preserved resting BRS (Tables 3 and 5). Overall mean % change in SBP ($-5.4\% \pm 0.6\%$ in group 1 vs. $-7.3\% \pm 0.6\%$ in group 2, $P>0.05$, Figure 1) was unchanged, but those with impaired BRS had a greater % change in DBP ($2.3\% \pm 0.5\%$ in group 1 vs. $-3.2\% \pm 0.4\%$ in group 2, $P<0.001$, Figure 1) and therefore the % reduction in MAP ($-2.3\% \pm 0.5\%$ in group 1 vs. $-6.8\% \pm 0.5\%$ in group 2, $P<0.01$, Figure 1) was greater in the group with preserved BRS. A smaller increase in HR was noted in those with impaired BRS ($0.9\% \pm 0.5\%$ in group 1 vs. $4.7\% \pm 0.7\%$ in group 2, $P<0.001$, Figure 1).

Important differences in the hemodynamic consequences of HD were noted. In particular, % change in CO ($-21.8\% \pm 1.1\%$ in group 1 vs. $-10.0\% \pm 0.7\%$ in group 2, $P<0.001$, Figure 2) and SV ($-22.2\% \pm 1.1\%$ in group 1 vs. $-13.4\% \pm 0.8\%$ in group 2, $P<0.001$, Figure 2) were both significantly greater in the impaired BRS group. In IDH-resistant patients with impaired resting BRS, there was a greater pressor response (% change in TPR; $42.4\% \pm 3.0\%$ in group 1 vs. $9.5\% \pm 1.1\%$ in group 2, $P<0.001$, Figure 2). There was no difference in % change

Table 2 Resting BP, resting BRS, IDH episodes during HD and RBV reduction for all patients separated according to IDH resistant with impaired resting BRS (group 1); IDH resistant and preserved resting BRS (group 2); IDH prone and impaired resting BRS (group 3) and IDH prone and preserved resting BRS (group 4)

	Overall	Stable (n=22)		Unstable (n=12)		P value
	(n=34)	Group 1 (n=10)	Group 2 (n=12)	Group 3 (n=6)	Group 4 (n=6)	
Geometric mean resting BRS (ms/mmHg)	5.78 (1.76–41.4)	3.36 (2.54–4.4)	8.80 (4.95–18.9)	2.63 (1.7–4.1)	13.48 (5.21–41.4)	0.001
Resting SBP (mmHg)	165.9 $\pm$ 8.0	179.0 $\pm$ 11.4	151.8 $\pm$ 13.1	179.5 $\pm$ 17.4	141.6 $\pm$ 21.4	0.237
Resting DBP (mmHg)	95.6 $\pm$ 5.9	105.3 $\pm$ 7.0	91.1 $\pm$ 8.1	95.8 $\pm$ 20.7	77.8 $\pm$ 11.4	0.438
Resting MAP (mmHg)	120.2 $\pm$ 6.5	131.4 $\pm$ 8.4	111.7 $\pm$ 9.9	125.3 $\pm$ 19.7	99.7 $\pm$ 14.1	0.326
Resting HR (beats/min)	74.5 $\pm$ 3.2	78.1 $\pm$ 4.2	68.1 $\pm$ 3.4	76.2 $\pm$ 3.66	73.8 $\pm$ 15.3	0.647
UF total (kg)	1.85 $\pm$ 0.13	2.29 $\pm$ 0.14	1.53 $\pm$ 0.23	1.38 $\pm$ 0.25	2.27 $\pm$ 0.2	0.01
Total IDH episodes (no.)	79	0	0	37	42	0.370
SBP <100 episodes (no.)	72	0	0	33	39	0.370
Symptomatic hypotension episodes (no.)	7	0	0	4	3	0.905
% RBV reduction	-6.74 ± 0.86	-7.47 ± 1.32	-5.43 ± 1.04	-5.78 ± 1.58	-9.13 ± 3.64	0.476

Data presented as mean $\pm$ SEM except BRS (GM with ranges).

BRS=baroreflex sensitivity; DBP=diastolic blood pressure; HR=heart rate; IDH=intradialytic hypotension; MAP=mean arterial pressure, RBV=relative blood volume; SBP=systolic blood pressure; UF=ultrafiltration.

Table 3 Percentage change in hemodynamic variables during HD in all patients separated according to IDH resistant with impaired resting BRS (group 1); IDH resistant and preserved resting BRS (group 2); IDH prone and impaired resting BRS (group 3) and IDH prone and preserved resting BRS (group 4)

	IDH-resistant patients		IDH-prone patients	
	Group 1	Group 2	Group 3	Group 4
% change in SBP	-5.4 ± 0.6	-7.3 ± 0.6	-17.4 ± 1.2	-5.4 ± 2.3
% change in DBP	2.3 ± 0.5	-3.2 ± 0.4	-6.6 ± 1.0	6.1 ± 1.2
% change in MAP	-2.3 ± 0.5	-6.8 ± 0.5	-13.8 ± 1.1	4.0 ± 1.3
% change in HR	0.9 ± 0.5	4.7 ± 0.7	8.1 ± 0.7	4.6 ± 1.4
% change in CO	-21.8 ± 1.1	-10.0 ± 0.7	-19.7 ± 1.1	-4.9 ± 1.3
% change in SV	-22.2 ± 1.1	-13.4 ± 0.8	-25.2 ± 1.2	-6.8 ± 1.5
% change in TPR	42.4 ± 3.0	9.5 ± 1.1	15.9 ± 2.1	10.6 ± 1.2
% change in BRS	83.2 ± 12.8	60.3 ± 8.5	-2.8 ± 5.9	-24.6 ± 4.0

BRS=baroreflex sensitivity; CO=cardiac output; DBP=diastolic blood pressure; HR=heart rate; IDH=intradialytic hypotension; MAP=mean arterial pressure; SBP=systolic blood pressure; SV=stroke volume; TPR=total peripheral resistance.

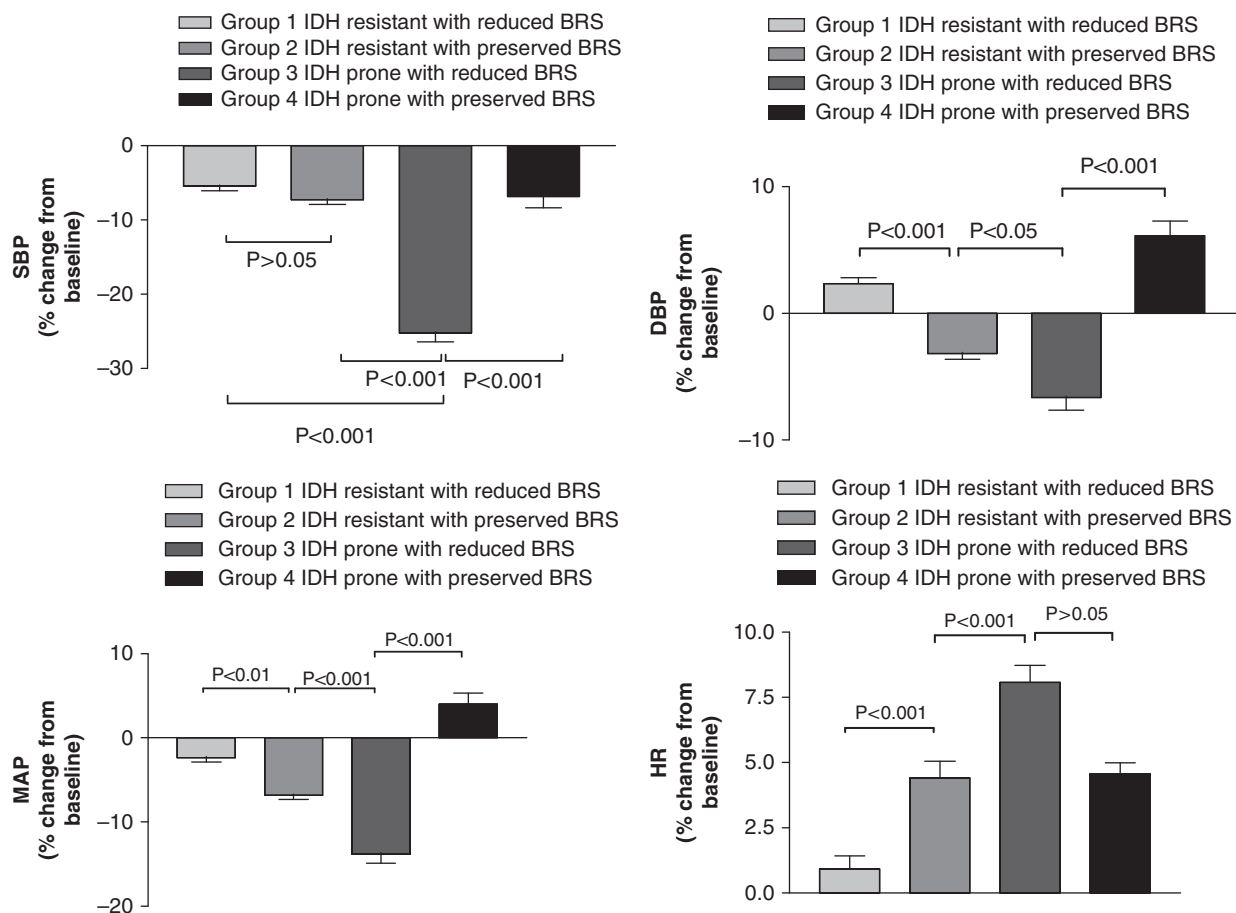


Figure 1 IDH prone patients with reduced BRS had the largest reduction in SBP, DBP and MAP. Conversely, this group had the largest increase in HR. DBP=diastolic blood pressure; HR=heart rate; IDH=intra-dialytic hypotension; MAP=mean arterial pressure; SBP=systolic blood pressure.

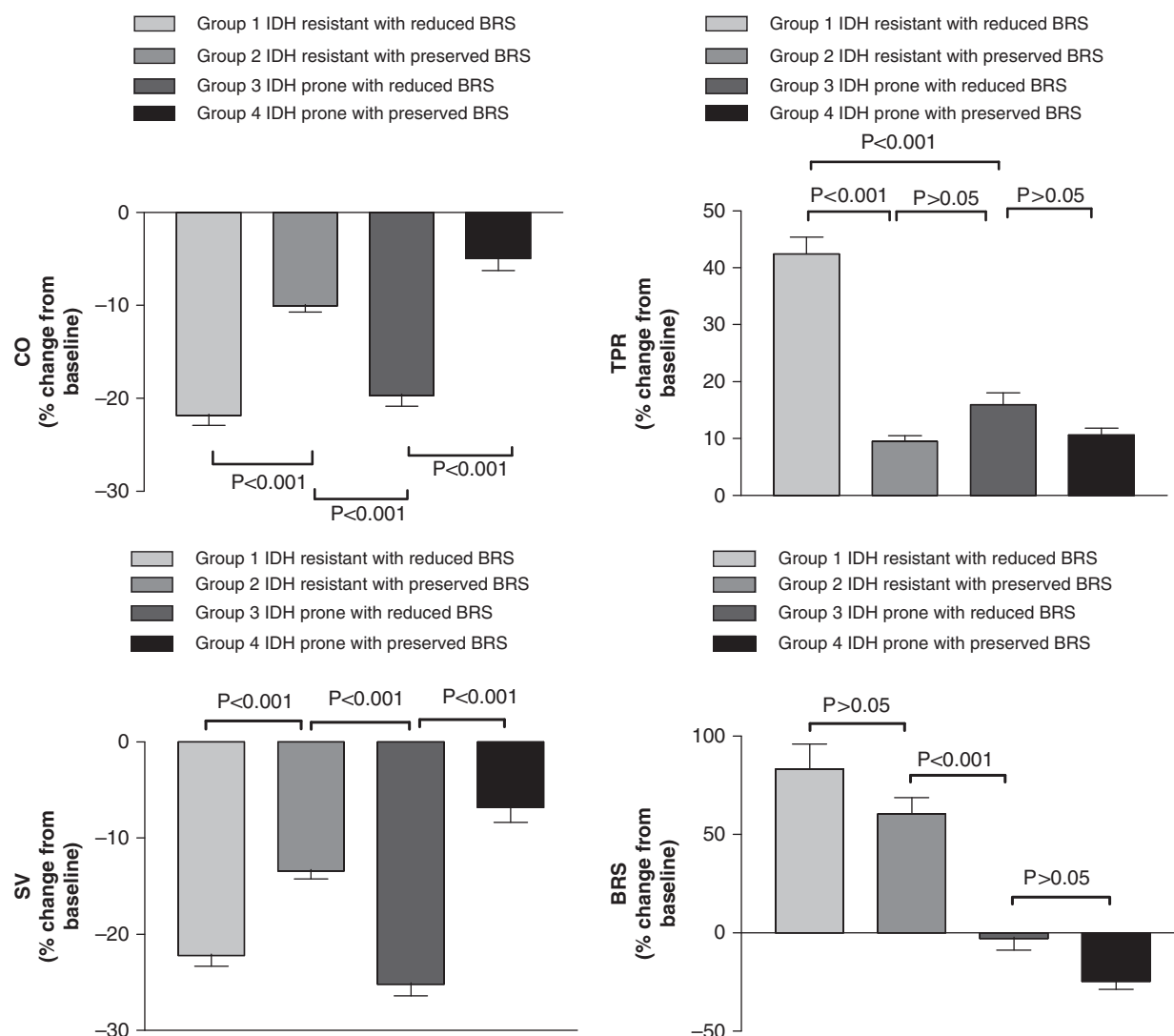


Figure 2 Percentage change in CO and SV was greatest in IDH-prone patients with impaired BRS. Percentage increase in TPR was greatest in IDH resistant patients with reduced BRS. Percentage change BRS differed according to hemodynamic stability. The greatest increase in BRS was seen in IDH resistant patients with reduced BRS. BRS=baroreflex sensitivity; CO=cardiac output; IDH=intradialytic hypotension; SV=stroke volume; TPR=total peripheral resistance.

in BRS between the IDH-resistant patients ($83.2\% \pm 12.8\%$ in group 1 vs. 60.3 ± 8.5 in group 2, $P > 0.05$, Figure 2).

Hemodynamic changes during HD in IDH-prone patients

Again, there were marked differences between the measured hemodynamic variables in the IDH-prone patients (Tables 3 and 5). Those with impaired resting BRS had the greatest reduction in SBP ($-17.4\% \pm 1.2\%$ in group 3 vs. $-5.4\% \pm 2.3\%$ in group 4, $P < 0.001$, Figure 1), DBP

($-6.6\% \pm 1.0\%$ in group 3 vs. $6.1\% \pm 1.2\%$ in group 4, $P < 0.001$, Figure 1) and MAP ($-13.8\% \pm 1.1\%$ in group 3 vs. $4.0\% \pm 1.3\%$ in group 4, $P < 0.001$, Figure 1). The greatest increase in HR was observed in those with reduced BRS ($8.1\% \pm 0.7\%$ in group 3 vs. $4.6\% \pm 1.4\%$ in group 4, $P < 0.001$, Figure 1). Furthermore, the reduction in CO and SV was greatest in those with reduced resting BRS (CO $-19.7\% \pm 1.1\%$ in group 3 vs. $-4.9\% \pm 1.3\%$ in group 4, $P < 0.001$, Figure 2; SV $-25.2\% \pm 1.2\%$ in group 3 vs. $-6.8\% \pm 1.5\%$ in group 4, $P < 0.001$, Figure 2). The pressor response (essential compensatory mechanism to

maintain BP during volume removal on HD) was similar in both groups (% increase in TPR $15.9\% \pm 2.1\%$ in group 3 vs. $10.6\% \pm 1.2\%$ in group 4, $P > 0.05$, Figure 2). The overall TPR response in the IDH-prone patients was not as marked as the TPR response from IDH-resistant impaired BRS patients. % change in BRS was similar in both groups ($-2.8\% \pm 5.89\%$ in group 3 vs. $-24.6\% \pm 4.0\%$ in group 4, $P > 0.05$, Figure 2).

Comparison of IDH-prone and IDH-resistant patients in their hemodynamic response to HD

Overall, IDH-resistant patients with impaired resting BRS had a large reduction in CO ($-21.8\% \pm 1.1\%$), and SV ($-22.2\% \pm 1.1\%$) and had the greatest increase in TPR ($42.4\% \pm 3.0\%$) (Figure 2). Furthermore, this group had the smallest increase in HR ($0.9\% \pm 0.5\%$, Figure 1). Conversely, IDH-prone patients with impaired resting BRS had a smaller reduction in CO ($-19.7\% \pm 1.1\%$), greater reduction in SV ($-25.2\% \pm 1.2\%$), a smaller increase in TPR ($15.9\% \pm 2.1\%$) but a greater increase in HR ($8.1\% \pm 0.7\%$) (Figures 1 and 2). Baroreflex sensitivity variability increased in IDH-resistant patients and decreased in unstable patients (Figure 2).

The relationship between percentage reduction in CO and RBV reduction at the end of HD

In IDH-resistant patients, there was no correlation between % reduction in RBV and % CO change (Table 4). However, in IDH-prone patients with impaired BRS, there was a strong correlation with volume removal and % reduction in CO ($r = 0.94$, $P = 0.006$, Figure 3, Table 4). Patients prone to IDH with reduced BRS were therefore dependent on maintenance of RBV to maintain CO

Table 4 Percentage change in cardiac output (CO) is dependent upon RBV reduction in those with reduced resting BRS and hemodynamic instability

	% RBV reduction vs. % change in CO	
	r	P value
Group 1	0.254	0.478
Group 2	-0.409	0.186
Group 3	0.935	0.006
Group 4	0.450	0.371

BRS=baroreflex sensitivity; CO=cardiac output; RBV=relative blood volume.

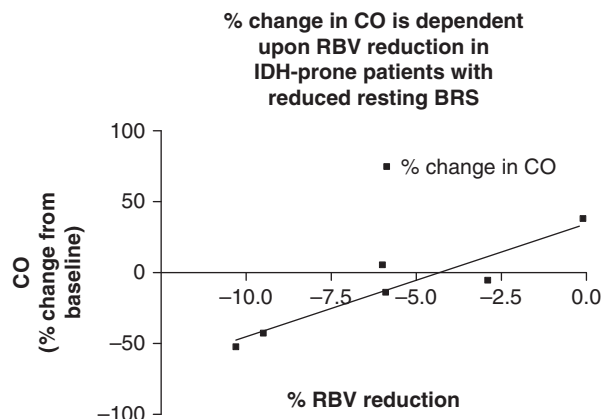


Figure 3 Percentage change in CO is dependent upon RBV reduction in those with reduced resting BRS and hemodynamic instability. BRS=baroreflex sensitivity; CO=cardiac output; RBV=relative blood volume.

during HD, whereas those with less impaired baroreflex function were not.

DISCUSSION

This study has demonstrated that even in patients prone to IDH, the presence of autonomic dysfunction is heterogeneous. The resulting variability in CV regulatory and functional failure contributes to relative hypotension. Furthermore, regardless of RBV reduction during HD, resting BRS appears to predict marked differences in hemodynamic response to HD. Lastly, IDH-resistant patients with impaired BRS exhibit the greatest increase in TPR during HD. A strong correlation exists between the effect of RBV reduction on CO at the end of HD but only in those unstable patients with impaired BRS, thereby rendering their CV response to HD highly volume dependent.

RBV monitoring and IDH

It has been proposed that IDH may be precipitated by severe hypovolaemia and that continual blood volume monitoring on dialysis is therefore critical to provide information on imminent IDH.¹⁶ However, Krepel et al. report considerable intra and interindividual variability in RBV reductions during HD ranging from -3.7% to -14.5% at 240 minutes.³¹ Barth et al. studied RBV reduction during HD in 60 unstable patients in order to identify an individual point in which RBV reduction would predict IDH episodes.³² However, because of the heterogeneous response to RBV, a critical point was not

Table 5 ANOVA comparison of hemodynamic variables during HD in all patients separated according to IDH resistant with impaired resting BRS (group 1); IDH resistant and preserved resting BRS (group 2); IDH prone and impaired resting BRS (group 3) and IDH prone and preserved resting BRS (group 4)

	% change in SBP	% change in DBP	% change in MAP	% change in CO	% change in TPR	% change in HR	% change in SV	% change in BRS
Group 1 vs. Group 2	>0.05	<0.001	<0.01	<0.001	<0.001	<0.001	<0.001	>0.05
Group 1 vs. Group 3	<0.001	<0.001	<0.001	>0.05	<0.001	<0.001	>0.05	<0.001
Group 1 vs. Group 4	<0.05	<0.01	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001
Group 2 vs. Group 3	<0.001	<0.05	<0.001	<0.001	>0.05	<0.001	<0.001	<0.001
Group 2 vs. Group 4	>0.05	<0.001	<0.001	<0.01	>0.05	>0.05	<0.001	<0.001
Group 3 vs. Group 4	<0.001	<0.001	<0.001	<0.001	>0.05	<0.001	<0.001	>0.05

BRS=baroreflex sensitivity; CO=cardiac output; DBP=diastolic blood pressure; HR=heart rate; IDH=intradialytic hypotension; MP=mean arterial pressure; SBP=systolic blood pressure; SV=stroke volume; TPR=total peripheral resistance.

reliably identified. Other weaknesses of RBV measurement include the influence of hydration status and an underestimate of blood volume changes due to the assumption that there is uniform mixing of red cells and plasma throughout the whole circulation.³³ Relative blood volume reduction is therefore important in the pathophysiology of IDH but as a single entity, is not sufficient to identify individuals at risk of IDH. In our study, RBV reduction was similar in all four groups, and as a result, the differences in hemodynamic response identified between the groups are not as a result of differences in RBV reduction.

Hemodynamic variables during HD

Hemodynamic changes during HD and ultrafiltration (UF) in a “typical” patient are relatively well defined. During UF, CO and SV decrease and MAP is maintained by an increase in TPR.^{34,35} Different groups have documented either a tachycardic, bradycardic or in some cases, unchanged HR responses during HD and UF.^{17,19,21,36} In our study, all groups demonstrated an increase in HR, but not necessarily a tachycardic response. The greatest response was seen in those IDH-prone patients with impaired resting BRS. The explanation for this is not clear. Hydration status before HD/UF has been shown to predict patterns in hemodynamic disturbance, and in some, but not all, overhydrated patients the HR response was sufficient to maintain BP stability.³⁷ Other explanations include the presence of unopposed cardiopulmonary reflex activation, thus resulting in a sympathetic response.

Our study has also demonstrated that the hemodynamic response to volume reduction in HD is indeed heterogeneous, and shown for the first time that at least part of the response can be predetermined by the measurement of resting BRS before HD.

Sympathetic activity and TPR response during HD

In agreement with other authors, our data in stable, IDH-resistant patients demonstrates a reduction in CO and SV during HD and an increase in TPR to maintain MAP. In addition, those with impaired resting BRS (group 1) in our study appear to have a greater increase in TPR than those with preserved autonomic function. This entity has been described before^{23,38,39} and may be due to background sympathetic overactivity, which is well recognized in the renal population,⁴⁰ and contributes to both hypertension and CV morbidity and mortality.⁴¹ The chronic elevation in SNS activity and associated reduction in adrenergic responsiveness, are the main determinants of impaired baroreflex buffering of arterial BP with aging.⁴² Reasons for sympathetic hyperactivity remain unclear but Cavalcanti et al. proposed that increased nitric oxide synthesis (also well recognized in CKD) inhibits responsiveness to adrenergic stimuli thereby inhibiting relationships between sympathetic activity and vascular response.⁴³ It is possible that these patients, in the absence of an intact BRS, rely upon the presence of cardiopulmonary reflexes to maximize sympathetic responses.⁴⁴ It might be argued that if BRS is impaired then the same disease processes must surely affect the cardiopulmonary reflex as well. However, we have previously demonstrated in HD patients that it is vascular calcification and other processes apparently restricted to the arterial tree that are primarily associated with reduced baroreceptor function.⁴⁵ The cardiopulmonary receptors existing within the low-pressure venous system, rather than the arterial tree, may indeed be spared these disease processes. Furthermore, there was a paradoxical rise in MAP in those IDH-prone patients with preserved resting BRS. Data from Cavalcanti et al. indicate that a rise in BP occurring during HD is as a

result of efferent reflexes being unbalanced in favor of the CP reflexes.⁴⁶

In IDH-prone patients with impaired resting BRS (group 3), hemodynamic changes appeared to be more marked than those with preserved BRS (group 4). In agreement with other authors,³⁶ we hypothesize that because the increase in TPR in IDH-prone group 3 patients was less than the response in IDH-resistant group 1 patients, this compensatory mechanism may be impaired and therefore insufficient to maintain MAP in IDH-prone patients. Seals and Dinunno concluded that tonic increase in sympathetic nervous system (SNS) activity causes agonist-promoted downregulation of β -adrenergic receptors, desensitization of the β -adrenergic signaling pathway and reduced vascular and cardiac tissue responsiveness to acute β -adrenergic stimulation.⁴² Other contributing factors to the reduced hemodynamic response include impaired myocardial contractility (an inappropriate greater reduction in CO) and HD-induced myocardial stunning.¹²

BRS during HD

As demonstrated by other authors, our data demonstrate that the presence of autonomic dysfunction is heterogeneous in the HD population, even in those prone to IDH. Scant data exist regarding the changes in BRS during HD, partly due to the traditional challenges of BRS measurement during physiologically changing environments. Published work, using HR variability methods, concludes that IDH-resistant patients exhibit increased sympathetic activity during HD.^{17,19,23} Our IDH-resistant population also demonstrates increased sympathetic activity, principally measured by surrogate markers, e.g., TPR, but also exhibits increased BRS variability (Figure 2). Increased BRS variability is associated with reduced BP variability and is physiologically advantageous, allowing adaptation to changing environments. In addition, reduced BRS variability has been associated with instability and sudden cardiac death.⁴⁷

RBV reduction and change in CO

Our data demonstrate that there was a strong correlation between percentage change in CO and RBV reduction at 240 minutes in IDH-prone patients with impaired resting BRS. This relationship was not evident in any other group. These patients also have a limited ability to increase BRS variability with hemodynamic stress. This group of patients appears to have a reduction in cardiac contractility, directly modulated by RBV reduction and are

therefore the most sensitive to a failure to maintain circulating volume. Stewart and colleagues reported that BP correlated with CO and end-diastolic volume, thereby supporting the hypothesis that there is a clear relationship between BP and CO which is amplified by increased peripheral resistance.⁴⁸ Furthermore, baroreflex regulation of CO is determined by myocardial contractility and venous return. An acute increase in carotid compliance attenuates carotid baroreflex control of CO and arterial BP in animal studies.⁴⁹ The authors later confirmed that the reduction in BRS mediated control of CO in hypertension appeared to be mediated by an increase in the resistance to venous return, which results in a decrease in venous return and thus a reduction in CO.⁵⁰ In addition, atrial deafferentation (involving removal of areas densely populated by atrial baroreceptors) in humans following cardiac transplantation impairs reflex vasomotor control of the circulation in response to low-level lower body negative pressure.⁵¹ Therefore, the baroreflex is not only of integral importance in the short-term regulation but is also essential in the control of the constituents of BP, the CO and TPR.

In summary therefore, measurement of resting BRS and sympathetic response, in particular TPR, may enable categorization of the hemodynamic response to RBV reduction during HD. For example, IDH-resistant patients appear to rely on a maximal TPR response and greater BRS variability during reductions in CO to maintain MAP. Conversely, IDH-prone patients appear to have a suboptimal TPR response, are less equipped to increase BRS variability, and possibly rely on an inadequate tachycardic response to attempt to maintain MAP during RBV reduction. Therefore patients with a reduced pressor response to volume removal and impaired resting BRS are those patients likely to experience recurrent cardiac injury through repeated myocardial stunning during HD and are therefore those that should avoid HD and receive peritoneal dialysis as the first choice of dialysis modality. Alternatively, nocturnal HD (6 $\times$ 8 hr/week) increases BRS and may indeed become increasingly become important in the management of these cardiovascularly fragile patients. Conversely, patients with an adequate peripheral pressor response despite impaired resting BRS may indeed be able to maintain hemodynamic stability during HD.

CONCLUSION

In conclusion, vasomotor instability, particularly in those patients receiving chronic HD, is an issue of increasing importance. The pathophysiology of IDH remains com-

plex and HD patients are increasingly well recognized to be heterogenous in their hemodynamic response to HD with UF. Autonomic dysfunction is important in the pathogenesis of the complex hemodynamic abnormalities that may exist in CKD patients. This is particularly pertinent given an aging and increasingly diabetic dialysis population. Relative blood volume monitoring has not been proven to predict HD tolerability in all HD patients. Individualizing HD therapy to prevent the development of HD complications remains an elusive goal, yet factors such as impaired resting BRS (which is dependent upon a number of factors including vascular calcification) and a suboptimal sympathetic TPR response, appear to predict those patients most likely to undergo hemodynamic instability. Because reduced BRS is potentially amenable to alterations in therapy, further investigation into the significance of autonomic dysfunction in vasomotor instability remains of prime clinical significance. Such extended assessment of patients may allow early identification of patients prone to IDH, allow selection of an optimal individualized intervention and potentially impact on the longer term CV outcomes.

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